

Sammy Sahtout

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EDUCATION

University of California, Davis

May 2028

B.S. in Electrical Engineering. GPA: 3.89

Relevant Coursework: Digital Logic Design, Circuits II, Device Physics, Embedded Systems, Electromagnetics

WORK EXPERIENCE

Project Engineer Intern

Sacramento, CA

Neural Lab

June 2026 – Present

- Junior developer for an ML-based touchless gesture recognition application using webcam-based computer vision.
- Resolved a performance issue in the gesture-recognition interface by tuning motion sensitivity parameters in the C++ source files, producing smoother and more precise tracking; validated and deployed to production on both Windows and macOS.
- Designed and implemented a Raspberry Pi USB bridge for medical devices, configuring the Pi in USB gadget mode to bridge Ethernet and USB connections and enable the workstation to recognize it as a USB device.
- Configured and maintained the software build environment using GitLab, CMake, and command-line tools; supported debugging and feature implementation.

Business Owner

El Dorado Hills, CA

Detailies (TikTok Shop)

August 2024 – January 2025

- Founded and operated an e-commerce business specializing in clothing and daily needs, fulfilling 500+ orders.
- Generated \$16,000+ in revenue; used Microsoft Excel to analyze sales trends, optimize inventory turnover, and track customer engagement metrics.

Project Management Intern

Sacramento, CA

Salam Center

June 2024 – July 2024

- Led a 30+ person project team, delivering a 5-minute execution plan and resolving role conflicts by reassigning responsibilities to keep the project on track.
- Applied Lean Six Sigma principles to improve project workflows and team efficiency.

PROJECTS

Steering Wheel Hand Detection System

AvenueE, UC Davis

- Built a capacitive hand-detection system as part of a three-person team using a self-fabricated aluminum-foil sensor, Arduino, resistor, passive buzzer, and LCD to detect when hands leave a simulated steering wheel.
- Converted the validated prototype into a custom KiCad PCB and improved system reliability by simplifying wiring, color-coding connections, and optimizing Arduino code to reduce detection lag.

Fire Detection Module

- Designed and tested a custom PCB in KiCad integrating a microcontroller with smoke, temperature, and CO₂ sensors for environmental monitoring and fire detection.
- Developed an ESP32-based distributed fire-detection system using smoke, temperature, and CO₂ sensors; triggered a Jetson Nano camera for visual verification before issuing alerts.

Solar Panel Robot Cleaner

- Engineered and prototyped a rail-based solar-panel cleaning robot in AutoCAD, building a scaled-down model designed for easy installation, manufacturability, and direct consumer use rather than a service-based model.
- Programmed an Arduino-based system using motors and proximity sensors to control brushing and bidirectional movement.
- Resolved a motor-driver power issue by integrating an external breadboard power-supply module for stable current; evaluated rain detection and a spray nozzle for water-assisted cleaning.

SKILLS

Technical Skills: Circuit Analysis, Embedded Systems, PCB Design, Digital Logic Design, Electromagnetics, USB/HID

Programming: C++, Python, Verilog, Java

Hardware: Oscilloscope, Multimeter, Signal Generator, Raspberry Pi, Arduino

Software Tools: KiCad, AutoCAD, GitLab, Linux, Intel Quartus Prime, Microsoft 365